Summary Table – Logistic Regression, KNN, Random Forest, SVM

Model	Accuracy	Recall (0)	Recall (1)	Precis
Logistic Regression	0.97	1	0.96	0.
K-Nearest Neighbors (KNN)	0.99	1	0.98	0.
Random Forest	0.98	0.98	0.99	0.
SVM	0.6	0	1	

Notes:

1. **Accuracy** = overall performance of the model on test data
2. **Recall (0/1)** = ability to correctly detect class 0 or 1
3. **Precision (0/1)** = correctness of predictions for class 0 or 1
4. **Correct Predictions** = True Positives for each class
5. **Support** = number of actual samples in each class in the test set

Summary of all the formulas you need for evaluating **classification models** like Logistic Regression, KNN, Random Forest, and SVM:

1.Confusion Matrix Components

Component	Meaning
TP (True Positive)	Correctly predicted positive class (1 predicted as 1)
TN (True Negative)	Correctly predicted negative class (0 predicted as 0)
FP (False Positive)	Incorrectly predicted positive (0 predicted as 1)
FN (False Negative)	Incorrectly predicted negative (1 predicted as 0)

- **Accuracy**

Measures overall correctness of the model.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **Precision**

Measures correctness of positive predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- Precision for **class 0** → TP0 / (TP0 + FP0)
- Precision for **class 1** → TP1 / (TP1 + FP1)

- **Recall (Sensitivity / True Positive Rate)**

Measures how many actual positives were correctly predicted.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- Recall for **class 0** → TP0 / (TP0 + FN0)
- Recall for **class 1** → TP1 / (TP1 + FN1)

- **F1-Score**

Harmonic mean of precision and recall.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Useful when you want a balance between precision and recall.

- **Support**

- Number of actual samples for each class in the dataset.

Example: Class 0 → 32, Class 1 → 48

Summary table and the performance metrics, we can calculate the **confusion matrix components**: **TP, TN, FP, FN** for each class.

Model **by model**.

Logistic Regression

From table:

- Test set count = 80
- Support: 0 → 32, 1 → 48
- Correct predictions = TP0 = 32, TP1 = 47

Formulas & Calculations:

1. **TP0 (True Positive for class 0)** = correct predictions for class 0 = 32
2. **FN0 (False Negative for class 0)** = actual 0 but predicted 1 = support0 - TP0 = 32 - 32 = 0
3. **TP1 (True Positive for class 1)** = correct predictions for class 1 = 47
4. **FN1 (False Negative for class 1)** = support1 - TP1 = 48 - 47 = 1
5. **FP0 (False Positive for class 0)** = predicted 0 but actual 1 = FN1 = 1
6. **FP1 (False Positive for class 1)** = predicted 1 but actual 0 = FN0 = 0
7. **TN (True Negative)** = total - TP - FP - FN (can also be derived from matrix layout)

Confusion Matrix:

	Pred 0	**Pred 1**
Actual 0	TP0 = 32	FN0 = 0
Actual 1	FP0 = 1	TP1 = 47

K-Nearest Neighbors (KNN)

- **Overall test set count:** 80
- **Support:** 0 → 32, 1 → 48
- **Correct predictions:** TP0 = 32, TP1 = 46

Formulas & Calculations

- **TP0 (True Positive for class 0)** = correct predictions for class 0 = 32
- **FN0 (False Negative for class 0)** = actual 0 but predicted 1 = support0 − TP0 = 32 − 32 = 0

- **TP1 (True Positive for class 1)** = correct predictions for class 1 = 46
- **FN1 (False Negative for class 1)** = support1 − TP1 = 48 − 46 = 2
- **FP0 (False Positive for class 0)** = predicted 0 but actual 1 = FN1 = 2
- **FP1 (False Positive for class 1)** = predicted 1 but actual 0 = FN0 = 0
- **TN (True Negative)** = total − TP − FP − FN
    - TN0 = TP1 = 46
    - TN1 = TP0 = 32

Confusion Matrix

	Pred 0	**Pred 1**
Actual 0	TP0 = 32	FN0 = 0
Actual 1	FP0 = 2	TP1 = 46

Random Forest

Test set = 133, support = 0 → 51, 1 → 82

Correct predictions: TP0 = 50, TP1 = 81

FN0 = 51 - 50 = 1

FN1 = 82 - 81 = 1

FP0 = FN1 = 1

FP1 = FN0 = 1

Confusion matrix:

	Pred 0	**Pred 1**
Actual 0	50	1
Actual 1	1	81

SVM

Test set = 80, support = 0 → 32, 1 → 48

Correct predictions: TP0 = 0, TP1 = 48

FN0 = 32 - 0 = 32

FN1 = 48 - 48 = 0

FP0 = FN1 = 0

FP1 = FN0 = 32

Confusion matrix:

	Pred 0	**Pred 1**
Actual 0	0	32
Actual 1	0	48

Summary Table – Confusion Matrix Values

Model	**TP0**
Logistic Regression	32
KNN	32
Random Forest	50
SVM	0
Note: TN can be calculated if needed as TN = total - TP - FP - FN.	

Based on the performance metrics from your experiments

Final Model Selection: K-Nearest Neighbors (KNN)

After comparing all models — Logistic Regression, Random Forest, SVM, and KNN — the **K-Nearest Neighbors (KNN)** model emerged as the **best model** for this dataset due to the following reasons:

1. **Highest Accuracy:** 0.99 → almost all predictions are correct.
2. **Balanced Recall:**
   - Class 0 recall = 1.00 → no class 0 instances missed
   - Class 1 recall = 0.98 → very few class 1 instances missed
3. **Strong Precision:**
   - Class 0 precision = 0.97
   - Class 1 precision = 1.00 → very reliable predictions
4. **Correct Predictions:** TP0 = 32, TP1 = 47 out of 80 → only 1 misclassification.

5. **Robust Overall Performance:** Handles both classes well with minimal errors, outperforming Logistic Regression, Random Forest, and SVM for this dataset.

Conclusion

The final selected model for deployment is: K-Nearest Neighbors (KNN).

It will serve as the production-ready model for predicting the target variable with **high accuracy, precision, and recall**.